Bio-GO-SHIP FCM Discrete Collection Protocol

Version 1.1

Authors

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Protocol Revision Record

Version	Date Implemented	Description of Revisions
1.1	2022-02-02	Modified for GO-SHIP protocol
1.0		Initial Release

Acronyms, Abbreviations, Terminology

Term	Definition
FCM	Flow Cytometric analysis
Niskin bottle	Plastic cylindrical bottle used for collecting water samples at different depths.
CTD Niskin rosette water	Sampling water gathered at depth from the conductivity, temperature, depth (CTD) Niskin rosette
DI water	Deionized water from the Milli-Q system
GO-SHIP	Global Ocean Ship-Based Hydrographic Investigations Program
Bio-GO-SHIP	Biological GO-SHIP
PPE	Personal Protective Equipment

Overview

This protocol describes collection FCM samples from both CTD Niskin rosettes. Water is collected with pre-rinsed 50 mL brown centrifuge tubes and a subsample preserved in a 2 mL cryovial for freezing. The recommended volume for GO-SHIP FCM samples is 1.8 mL.

Supplies Required

Equipment:
Rinsed 50 mL centrifuge tubes
P1000 pipettor
P100 pipettor
Small cooler
4°C refrigerator
–80°C freezer
Vented fume hood
PPE (lab coat, eye protection, and gloves)

Consumables:
2 mL Cryovials (internal thread)
1 mL pipet tips with filter
20 uL pipet tips with filter
Pre-printed tube labels
Nitrile gloves

Chemicals:
EM Grade 25% glutaraldehyde (10 mL glass ampule)
Kolliphor P188 (Pluronic 2% solution) – note 2% solution is created using Deionized water and is autoclaved then dispensed (in sterile hood) and stored in 900 mL aliquots. Store the aliquots of Pluronic solution in the freezer until use.

Preparation of Preservation solution

To prepare the Preservation Mix (working solution) mix the Glutaraldehyde (25%) and Kolliphor P188(2%) 50/50.

1. Thaw 900uL Kolliphor (2% Pluronic solution)
2. Turn on the hood
3. In the hood wearing PPE, add 900 microliters of 25% EM Grade glutaraldehyde to the vial of 900 microliters Kolliphor.
4. Mix well. This mixture (total volume - 1.8 mL) is the **Preservation Mix**. Store preservation mix in refrigerator between casts or inline sampling. Only prepare the Preservation Mix as needed.

NOTE: 25% Glutaraldehyde **Please use caution, and please wear gloves and always work in the fume hood! Store dispensed glutaraldehyde in the refrigerator until use** (be sure it is tightly sealed before removing from the hood). All waste that comes in contact with the glutaraldehyde (i.e. pipette tips for dispensing preservation mixture and contaminated gloves) should be stored in the fume hood in the waste ziploc bag taped to the side.

Collection protocol

1. Collect a sample from either the inline or CTD cast (8 depths) into an appropriately labeled rinsed dark/brown centrifuge 50 mL tube. Ideally store the sample tube(s) in a small cooler. When filling the tubes with sample water - rinse with the inline or CTD water three times before sample collection.
2. Using a marker, prior to adding the cryolabel (see step 3) label each cryovial with the sequential number used in the spreadsheet log. Have a separate log for the inline (transect) samples and the CTD samples.
3. Label 2 ml cryovials using the label maker protocol.

Format of CTD BioCAST Label:	Format for Inline sample:
CruiseX CTD XXX	Cruise X TXXX
YYYY-MM-DD N=XX	YYYY-MM-DD UTC Time

4. In a hood, add 18 microliters of the preservation mixture (10 microliters/ml) using the P100 pipette. Use the same pipette tip repeatedly (for the preservation mixture) **to add the preservation mix to each tube.**
5. Fill each labeled 2 ml cryovial with 1.8 mL of sample water (dispense 900 microliters x2). Keep them in the dark (you can use foil).
6. Mix each sample well by inverting several times.
7. Wait a minimum of 10 minutes (use a timer).
8. Flash freeze in liquid nitrogen (preferred) or at -80C. Use separate boxes for CTD casts and Inline samples. Be sure to label the starting and end vials on the sides of the boxes. Order them sequentially in the box.
9. After the preservation is complete, fill out the digital logs to record the metadata associated with each sample.

Sampling Scheme (with one daily bio cast)

CTD rosette sampling at ~12 pm at eight depths (5, 40, 75, 100, 150, 200, 500 and 1000 m)
Duplicate sample(s) taken every 3-4 days.